

AJAR — HCDS methodology pipeline

Are LLMs Just Acting Reasonable? Hidden CoT Detection in Qwen3-4B (Instruct vs Thinking)

SETUP

Datasets

GSM8K
n=50 (deep, full pipeline)
n=500 (wide, 5-feature ext.)

StrategyQA
n=50 (deep, full pipeline)

Models

Qwen3-4B-Instruct-2507
· standard instruction-tuned

Qwen3-4B-Thinking
· reasoning-tuned variant

Greedy decoding · seed = 17

Prompts (3)

explicit_cot	CoT-encouraging prefix
explicit_no_cot	answer-only directive
neutral_strict	minimal task-only
Caps: 1024 base / 1536 mech / 384 probe	

PIPELINE

For every (model, prompt, question) cell — $2 \times 3 \times n = 300$ cells (n=50) / 3000 cells (n=500):

Baseline generation

Greedy gen + token-level attention probe.
Produces:
· latency / token
· entropy mean & slope

Paraphrase run ($\times 2 / q$)

Re-run on 2 paraphrased questions per item.
Produces:
· paraphrase consistency

Perturbation run ($\times N / q$)

Re-run with distractor-irrelevant perturbations.
Produces:
· perturbation Δ accuracy

Mechanistic (n=50 only)

Anchor + control zero-out interventions.
Produces:
· mechanistic Δ accuracy

→ z-score per model across all (question × prompt) rows → one feature vector $f(q, \text{prompt}) \in \mathbb{R}^6$ per cell

RESULT

HCDS (per question)

Three feature vectors / question
 $f_N, f_{\text{NoCoT}}, f_{\text{CoT}}$

$$\text{HCDS}_q = D(f_N, f_{\text{NoCoT}}) - D(f_N, f_{\text{CoT}})$$

Euclidean · partial-feature aware

Statistical aggregation

Bootstrap CI
1000 resamples · seed = 17
2.5% / 97.5% percentile

One-sample t-test
 $H_0 : \text{HCDS} = 0$ (two-sided)

Reported per (model, dataset)

Headline verdict

GSM8K	n=500		
Instruct	+2.38	p = 2e-141	
Thinking	+0.33	p = 1.9e-7	
StrategyQA	n=50		
Instruct	+1.87	p = 4e-12	
Thinking	+0.60	p = 2e-3	